

Figure 1 is a scatter plot showing the relationship between Predicted $\text{PM}_{2.5}$ ($\mu\text{g m}^{-3}$) on the y-axis and Observed $\text{PM}_{2.5}$ ($\mu\text{g m}^{-3}$) on the x-axis. The plot includes a dashed 1:1 line and a solid green calibration line. The data points are blue circles. Marginal histograms are shown on the top (for Observed) and right (for Predicted) axes. Performance metrics are displayed in a box in the upper left corner of the plot area:

- $R^2 = 0.732$
- $\text{RMSE} = 5.58 \mu\text{g m}^{-3}$
- $\text{MAE} = 4.06 \mu\text{g m}^{-3}$
- $\text{Bias} = -0.03 \mu\text{g m}^{-3}$

A legend in the bottom right corner identifies the dashed line as the '1:1 line' and the solid green line as the 'Calibration line'.

b)

Residual (predicted – observed, $\mu\text{g m}^{-3}$)

Observed $\text{PM}_{2.5}$ range ($\mu\text{g m}^{-3}$)

Mean bias

≤ 20 (n=571)

20–35 (n=110)

> 35 (n=39)

Detailed description: This box plot displays the distribution of residuals (predicted minus observed) for three observed PM2.5 ranges. The y-axis represents the residual in $\mu\text{g m}^{-3}$, ranging from -30 to 30. The x-axis shows the observed PM2.5 ranges: ≤ 20 (n=571), 20–35 (n=110), and > 35 (n=39). Each box plot includes a median line, a box representing the interquartile range, and whiskers extending to the range of the data. Individual data points are overlaid on each box. A purple diamond indicates the mean bias for each group. The legend shows a purple diamond for 'Mean bias'.

Observed $\text{PM}_{2.5}$ range ($\mu\text{g m}^{-3}$)	n	Median	Q1	Q3	Min	Max	Mean bias
≤ 20	571	~0.5	~-1.5	~3.0	~-9.0	~22.0	~0.5
20–35	110	~-3.5	~-8.0	~2.0	~-18.0	~20.0	~-2.5
> 35	39	~-5.0	~-12.0	~2.0	~-27.0	~14.0	~-5.0